

Isaiah Gangadeen

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Professional Profile

An Applied Economist and Lecturer in the Department of Economics at the University of Guyana, specializing in industrial organization, energy/resource economics, environmental economics, and development. Research is grounded in microeconomic analysis, with particular interests in how firms, markets, institutions, and technology shape innovation, resource allocation, environmental impacts and development transitions. Work employs quantitative and computational methods to applied economic questions, including causal inference, optimization, economic modelling, and policy simulation. Experience includes firm and industry analysis, industrial competitiveness, and macroeconomic and policy analysis.

Publications

Google Scholar: <https://scholar.google.com/citations?user=2GLnTYMAAAAJ&hl=en>

Peer-Reviewed Journal Articles:

- 2025 Gangadeen, I., Shah, KU. Dynamic connectedness of corn, plastic and energy markets: Understanding Bioplastic Competitiveness, <https://doi.org/10.1016/j.jclepro.2025.145256>
- 2023 Shah, KU., Gangadeen, I., Integrating bioplastics into the US plastics supply chain: towards a policy research agenda for the bioplastic transition, *Frontiers in Environmental Science*, 11, 1245846 <https://doi.org/10.3389/fenvs.2023.1245846>

Manuscripts Under Review:

- Gangadeen, I., Shah, KU. On the Limits of Polymer Substitution Sustainability when considering Carbon Leakage and Firm-Level Decision-Making
- *Journal of Cleaner Production*
- Gangadeen, I. Oil Production under Production Sharing Contracts: Policies, Constraints and Outcomes in Guyana
- *Energy Economics*
- Gangadeen, I., Shah, KU. Resource Curse and Emissions Pathways: An Emissions Decomposition of the Effects of Oil Production on Tropical Oil Producing Countries
- *Journal of Cleaner Production*

Education

- **Ph.D. Energy and Environmental Policy** - University of Delaware 2026
Thesis: "Oil Wealth in a Carbon Sink: Rethinking Development and Environmental Compatibility."
Awarded: Most outstanding for its theoretical formulation or empiricism.
 - **M.Sc. Development Economics** - University of Manchester 2020
 - **B.Sc. Economics** - University of Guyana 2016
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Professional Appointments

- **Lecturer** 2021 – Present
Department of Economics, University of Guyana, Guyana
Courses Taught:
 - ECN 4106 – Economics of Innovation
 - ECN 3103 – Industrial Organization
 - ECN 2201 – Introductory Econometrics
 - ECN 1101 – Introductory Mathematics
 - ECN 1100 – Introductory Microeconomics
 - ECN 1200 – Introductory Macroeconomics

- **Research Associate** 2022 – 2026
Island Policy Lab, Center for Energy and Environmental Policy, University of Delaware, USA
 - Selected Projects:
 - Established Program to Stimulate Competitive Research (EPSCoR) Renewable & Recyclable Polymers (R2P) tasked with the integration of Bioplastics and Plastic Recycling technologies into the US plastic industry.
 - The IPL-IDB Caribbean [Climate Governance Toolkit](#).
 - UN and American Chamber of Commerce in Trinidad and Tobago (AMCHAMTT) report on the ESG Landscape of in Trinidad and Tobago. Summary report: [AMCHAMTT Quarterly Online Magazine](#).
 - Teaching Responsibilities:
 - ENEP 425/625 – Energy Policy and Administration
 - ENEP 426 – Climate Change Policy
 - ENEP 485 – Senior Seminar on Energy & Environmental Policy

- **Economist** 2018 – 2021
Business Strategy and Policy Unit, Ministry of Tourism, Industry and Commerce, Guyana

- **Economic and Financial Analyst** 2016 – 2017
Fiscal and Monetary Division, Ministry of Finance, Guyana

Skills

- **Quantitative Methods:** Econometrics, causal inferences, policy analysis, analytical and computational methods, and policy simulations.
 - **Programming and Statistical Software:** R, Python, Stata, EViews, SPSS.
 - **Data Analysis and Visualization:** Power BI, Tableau.
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Honors and Awards

- **2026 Marvin B. Sussman Prize** issued by Joseph R. Biden, Jr. School of Public Policy & Administration. Awarded to a Ph.D. graduate whose dissertation is judged to be the most outstanding in its theoretical formulation or empiricism.
- **2025 Poster Competition Runner-Up** issued by the United States Association for Energy Economics (USAEE) North American Conference
- **2024 Excellence in Research and Innovation** issued by University of Delaware Black Graduate Student Association

Professional Training

- Monitoring and Evaluation 2016 - 2017
- Debt Management 2016
- Macro-Fiscal Management 2016

Presentations

- 2025 Resource Curse and Emissions Pathways: An Emissions Decomposition of the Effects of Oil Production on Tropical Oil Producing Countries at *the 42nd United States Association for Energy Economics (USAEE) / International Association for Energy Economics (IAEE) North American Conference*
 - 2025 *Second Regional Workshop on Building a Science and Data-based Agenda for Decision-making on Risk and Resilience in the Caribbean*
 - 2023 *Panelist on “International Climate Policy with a Caribbean Lens” at the [Climate Gen Z Conference](#) hosted by the Caribbean Environmental Science and Renewable Energy (CESaRE) Journal.*
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